Engineering ML project: energy use at a steel plant

CML3112 | Civil Engineering and Architecture | Part A 30 marks + Part B 15 marks = 45 marks, scaled to 40 percent of the course | Online, 3 hours, individual work, marked by computer

At a glance

The window: opens Monday 23 November 2026, 00:00 and closes Wednesday 25 November 2026, 23:59 Kampala time, the same for all three courses. You have 3 hours from the moment you press Start test. Start by Wednesday 25 November 2026, 20:59 to get the full 3 hours.

Online, from anywhere: your own laptop, your own Google account and an internet connection. There is no room and nobody watching. You use the same entry page, countdown and Submit test steps as Practical Test 1.

The rehearsal (unmarked): released Tuesday 17 November 2026, right after Class 3. Same structure and the same kinds of task, on the same file with a different target. A solved walk-through is not given.

One notebook, one person. Everything is marked automatically. Part A is the coded notebook (30 marks). Part B is the decision memo (15 marks), written as a form inside the notebook.

It covers only what we taught in Weeks 1 to 10 (up to Class 3). There is nothing new to learn.

Every student works on the same real file, so your numbers will match a classmate's. Most marks come from checks of your functions on rows you cannot see, and a short online understanding check may follow for anyone.

1. The situation and the two questions

You work for the energy manager of a steel plant (DAEWOO Steel, Gwangyang, South Korea). They give you one year of electricity readings: 35,040 rows, one every 15 minutes for 2018. Each row has the energy used (Usage_kWh, the target variable), the lagging and leading reactive power, two power factors, the seconds since midnight (NSM), the day and a CO2 column. You answer two questions.

	What you must answer
Question 1 **How much?**	How much energy will the plant use in a 15-minute reading, in kWh, in months the model has never seen? Compare a baseline model, a multiple linear regression model and a decision tree of depth 4, on the same rows, with MAE and RMSE in kWh. The features are the lagging and leading reactive power, NSM and a weekend flag.
Question 2 **Warn or not?**	A reading is a warning (1, the positive class) when its Usage_kWh is at or above 60 kWh, and no warning (0) below it. A quarter hour at or above 60 kWh means at least 240 kW on average, which is when peak-demand charges build up. Give each reading a probability with logistic regression, choose a decision threshold, and report the confusion matrix counts, precision, recall, F1, the cost and the ROC AUC. A false negative (a missed warning) costs 5 and a false positive (a false alarm) costs 1 (invented teaching costs).
The client's limits	A typical prediction error of at most 10 kWh (MAE) and a prediction time of at most 50 ms. A supplied rule in your notebook picks one model with these limits. Both numbers, and the 60 kWh limit, are invented for teaching.

2. The data and the test set

- **Where the data comes from.** Sathishkumar V E, Changsun Shin and Yongyun Cho (2021). Steel Industry Energy Consumption [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C52G8C. Licence CC BY 4.0. The file is stored inside the supplied setup cell of your notebook and written out with a check, so nothing is uploaded or downloaded.
- **What is really in it.** We checked the file before the project. It has no missing cells and no repeated rows. It does have three things you must handle in Task 1: each day's last reading is written with that day's date and the time 00:00 (it really belongs to midnight at the end of the day); one reading on 7 November has every

measurement 0 (the meter recorded nothing); and CO2(tCO2) is calculated from Usage_kWh, so it would leak the answer. The two power factors are calculated from the energy and the reactive power, so they are not used as features either.

- **Split by month.** Train on January to August (23,328 rows, the training set), validate on September and October (5,856 rows, the validation set). November and December (5,855 rows after cleaning) are the test set.
- **Why not a random split?** Readings 15 minutes apart are almost the same. A random split would put near-copies of the same hour in the training and validation sets, and let the model learn from days after the ones it is tested on. The score would look better than it will be next month. Splitting by month keeps the future in the future.
- **Cross-validation by month.** Each month from January to October takes a turn as the unseen month (the idea of GroupKFold, with the month as the group).

How the test set is opened once

The supplied cell "Open the test set" is in your notebook. Do not edit it. It defines open_test_set(model_name, threshold).

Near the end you call it ONE time: test_result = open_test_set(chosen_model, chosen_threshold), after you have chosen the model and the threshold on the validation set. It cleans the whole file with your clean_rows, trains your functions on January to October, evaluates on November and December and prints the result.

The notebook counts how often you call it. A second call loses the "opened once" mark, even with the same choice. A second call with a different model or threshold also prints a warning that is saved in your notebook and loses the test set marks.

Never use November or December to choose a model, a threshold or a hyperparameter. Once you have looked at them and then changed your mind, they can no longer tell you how the model behaves on new months.

3. How the window works

1. **Before you start.** Use a charged laptop, your own Google account and a working internet connection, somewhere quiet. Do the rehearsal first. Open the entry link or scan the QR code your lecturer shared. Check the course and the title "Engineering ML project".
2. **Start.** Enter your full name, your Google email and your Student Registration Number, then press Start test. Your three hours begin at that moment. Opening the page alone does not start the clock. Use the same Google account in Colab and in the form.
3. **Your notebook.** Press Open my test in Colab. It is your own private notebook; no copy is needed. Run the first code cell (your details and the countdown), then the supplied setup cell. Work from the top. Save often with File, then Save.
4. **The deadline.** Your deadline is the earlier of three hours after you started and the window's close (Wednesday 25 November 2026, 23:59). The clock cannot be paused. Refreshing, a new device or an internet interruption does not restart it. If you start after Wednesday 25 November 2026, 20:59, you only have the time left until the close.
5. **The end.** At your deadline the Submit test button switches itself off, the lecturer's server refuses new snapshots, and a form response after the deadline is kept for review and is not marked. Colab may still let you type; nothing you type after the deadline is marked.
6. **Help.** The lecturer answers questions about wording and technical problems on the class channel, 08:00 to 20:00 each day of the window (lecturer to confirm these hours). The lecturer does not write or debug your answers.
7. **If something fails.** Report it at once on the class channel with a screenshot (Colab, the entry page or the form). The lecturer can authorise one fresh attempt; it still ends at the window's close. Do not make a second identity and do not use another student's file. If Colab disconnects, reconnect and run the countdown cell and the setup cell again; your saved notebook is still in Drive and the timer continues.

Suggested time plan	Time
Setup: open your notebook, run the first cells	10 min

Suggested time plan	Time
Tasks 1 and 2: clean the rows, split by month	35 min
Task 3: the three models	30 min
Task 4: warnings and the cost table	25 min
Task 5: cross-validation by month	15 min
Tasks 6 and 7: choose a model, open the test set once	10 min
Task 8: the Part B memo form	25 min
Check, restart and run all, submit	20 min
Spare time (breaks, reconnecting)	10 min
Total	180 min

4. What you hand in, exactly

Item	Detail
One notebook	Your issued notebook, with Tasks 1 to 7 done and the Part B memo form (Task 8) filled in. Restarted, run from the top, then saved.
How	The last cell of the notebook. Run it, click Submit test, open the form link, check your name and registration number, and press Submit. Wait for the message that your response was recorded. Allow at least 20 minutes for this step.
Nothing else	No ZIP, CSV, PDF, Word file, screenshot or separate memo. One submission per person. A new submission before your deadline must be the complete notebook; the latest one on time is marked.

5. Allowed tools and the AI rule

These are the same rules as Practical Test 1.

- **Permitted:** Google Colab, your notebook, Python with pandas, NumPy, matplotlib and scikit-learn as used in class, the submission form, and your lecture slides, your lab notebooks from Classes 1 to 3, the revision guide, the cheatsheet and your own course notes.
- **No AI or outside help:** do not use Gemini, ChatGPT, Copilot or any other AI tool to generate, complete, rewrite, explain or debug your answers. Do not get answers from another person, collaborate, message or copy someone else's work.
- **Understanding check:** after the window, any student may be asked, in an online call, to explain their submitted code or make a small change to it, without AI or another person's help.
- **Declaration:** by submitting, you confirm that you completed this project yourself without AI assistance or another person's help, using only the permitted resources, and that you can explain and change the code you submitted.

The copy checks and the identity checks run on every notebook. The lecturer's computer compares notebooks for copying and lists unusual cases for a person to look at. A flag never changes a mark by itself.

6. Rules that make your notebook run every time

1. **Restart and run all.** Choose Runtime, then Restart session and run all, before you submit. Every cell must run from the top without an error, in under five minutes in total and under 30 seconds per cell. With 35,040 rows, look timestamps up in a set: a list is far too slow.
2. **Your own cells are fine.** You may add your own cells. Your work is marked wherever it is, as long as the required function names exist when the notebook runs from the top.

3. **Keep the supplied cells and names.** Do not edit the supplied cells (the ones that say Do not edit this cell). Keep each function name, argument order, returned keys and variable name exactly as printed. The test-set cell calls your functions by these names.
4. **Use only the arguments.** A function uses the lists it is given (and the constants such as TARGET and FEATURES), never a global table. It is marked on other rows you cannot see.
5. **Train on the training set only.** The imputation, the StandardScaler and the baseline model's mean are learned from the training rows. Using validation or test rows in them is data leakage and loses marks.
6. **Seeds.** Keep random_state=17 for the decision tree. Do not add random choices without a seed, and never use a random split.
7. **Calculate, do not type.** Your results come from your code. A number typed in by hand earns nothing in Part A.
8. **No extras.** No pip installs, no downloads, no extra files.

7. How Part A is marked (30 marks)

The lecturer's computer runs your notebook from the top in a locked box. The self-check cells in your notebook are not marked: they use tiny examples with known answers and show PASS or TODO. The hidden checks run your functions on other rows built from the same plant file and compare the numbers. You cannot see them, so test your functions with your own small examples too. Each item earns the share of its checks that pass, so one wrong task does not cost the others. Because everyone has the same file, your results on the file are worth only 1 mark (A8); the functions are worth 21; six marks are for how you worked (items A10 to A13).

Part A item	What is checked	Taught in	Marks
A1. Clean the rows	Your function on rows built from the real file with problems added: midnight readings moved to the next day, first copy of a timestamp kept, an empty reading, impossible values and a missing target dropped, CO2(tCO2) removed, a missing feature left as None, your input unchanged.	Weeks 3 to 7	3
A2. Split by month	Your function on other rows, also in a different order: months 1 to 8 training set, 9 and 10 validation set, 11 and 12 test set.	Week 10	2
A3. Baseline model, linear regression and decision tree	Baseline model, linear regression and decision tree predictions on other rows with missing values, fitted on training rows only, with the median imputation learned from training rows.	Weeks 8 and 9	4
A4. MAE and RMSE	MAE and RMSE on other numbers, including a case with no error.	Week 8	2
A5. Warning probabilities (logistic regression)	Warning probabilities (logistic regression) on other rows, for the limit you are given and for another limit.	Week 9	2
A6. Confusion matrix, precision, recall, F1, cost and AUC	Confusion matrix counts, precision, recall, F1, cost and AUC at four decision thresholds, including one where nothing is flagged.	Week 9	4
A7. Cross-validation by month	Ten month folds, their mean and their spread, on other rows (November and December must be left out).	Week 10	4
A8. Your results on the data	Your validation errors, cross-validation, AUC and chosen model on the file, wherever they are in your notebook (a variable of any name, or printed). Numbers typed in by hand earn nothing.	Weeks 8 to 10	1
A9. The test set result	The test set opened with your chosen model and threshold; the supplied cells unedited; the test result is the correct one for your choice.	Week 10	2

Part A item	What is checked	Taught in	Marks
A10. Test set opened once, after the choice	The test set opened exactly once, after you chose the model and the threshold, with that choice; no November or December row among your development rows.	Week 10	2
A11. No data leakage: preprocessing fitted on training data only	Four probes: the imputation, the baseline model, the StandardScaler and the cross-validation by month use only the rows they are given for training (no data leakage).	Weeks 8 to 10	2
A12. Reproducible with the seed	Calling your regression function twice gives the same answer; the decision tree keeps random_state=17; no unseeded randomness; your saved predictions equal a fresh run.	Weeks 8 to 10	1
A13. Charts and pipeline requirements	Chart 1 and Chart 2 (under any name) each have a title, named axes and the unit, and plot the right data; the required functions and the MEMO form exist.	Weeks 3 to 8	1
Total			30

8. Part B: the decision memo form (15 marks)

The memo is for the plant's energy manager, who is not a data scientist. You write it as a form: a Python dictionary called MEMO in Task 8. Every field is checked automatically, so type your numbers carefully (two decimals are enough, in kWh). The lists of limits and next steps are printed in your notebook; each course has its own lettering.

Field	What you enter and how it is checked	Marks
recommended_model, recommended_threshold	A model you trained ("baseline", "linear" or "tree") and a decision threshold (0.3, 0.5 or 0.7). They must be the choice your rule and cost table made.	3
validation_mae	The validation MAE of the three models. Compared with the values your own notebook computed.	3
test_mae, test_recall	The MAE and the warning recall on the test set from your one opening. Compared with the value the marking computer calculates from the same predictions.	2
cv_mean_mae	The mean MAE of your cross-validation by month. Compared with your own cv_summary.	1
limits	Three different letters: the statements that are true limits of this study. Ten statements are listed and six are false. One mark for each true one.	3
next_step	One letter: the best next step, from five.	1
answer_choice, answer_test_set	Two short written answers of 8 to 80 words each: why you recommend your model (use a validation number and say what you compared it with) and why the test set is opened only once. One mark each. Several correct wordings are accepted; write sentences in your own words.	2
Total		15

9. Rehearsal, help and results

- Rehearsal: Tuesday 17 November 2026, right after Class 3. It predicts the lagging reactive power (kVarh) from the same file, with its own warning. Do it with a timer. It is unmarked.
- Window: opens Monday 23 November 2026, 00:00 and closes Wednesday 25 November 2026, 23:59 Kampala time.
- Questions about task wording and technical problems: the class channel, 08:00 to 20:00 each day of the window (lecturer to confirm these hours).
- Results: marks are final about a week after the window closes, by Wednesday 2 December 2026.

- Practise the whole path in the rehearsal: restart and run all, fill in the MEMO form until the memo self-check shows 10 of 10, and read the self-check output.

Before you submit

☐	Identity cell correct (name, email, registration number). COHORT and TEST_ID unchanged.
☐	Restart session and run all: no error, every cell ran.
☐	The test set was opened one time, after the choice. Chart 1 and Chart 2 have a title, named axes and the unit.
☐	The MEMO form is complete and the memo self-check passes.
☐	File saved, Submit test clicked, form submitted, confirmation seen before your deadline.

Data credit: Sathishkumar V E, Changsun Shin and Yongyun Cho (2021). Steel Industry Energy Consumption [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C52G8C. Licence CC BY 4.0. Introductory paper: Efficient energy consumption prediction model for a data analytic-enabled industry building in a smart city, Building Research and Information 49(1), 127 to 143 (2021).